

Project Design Phase Proposed Solution Template

Date	06 July 2026
Team ID	Individual
Project Name	India's Agricultural Crop Production Analysis (1997-2021)
Maximum Marks	2 Marks

Proposed Solution Template:

Project team shall fill the following information in the proposed solution template.

S.No.	Parameter	Description
1.	Problem Statement (Problem to be solved)	India's crop production data across states, districts, seasons and years (1997-2021) is scattered and largely raw, making it difficult for farmers, policymakers and researchers to identify long-term trends, high/low-yield regions or crop performance patterns needed for informed agricultural planning and policy decisions.
2.	Idea / Solution description	An interactive Tableau dashboard that consolidates 25 years of India's crop production data (state, district, crop, season, area, production, yield) into trend lines, choropleth maps, comparative charts and calculated KPIs, letting users explore production patterns by state, crop, season and year, and quickly spot growing or declining crops/regions.
3.	Novelty / Uniqueness	Unlike static government reports or raw spreadsheets, this solution offers one unified, interactive platform with drill-down (state → district → crop → year), calculated yield/productivity metrics, and shareable public dashboards - usable without any coding or GIS expertise.
4.	Social Impact / Customer Satisfaction	Helps farmers make informed sowing decisions using area-specific crop performance data, helps policymakers identify underperforming regions for targeted support, and gives researchers and students a ready analytical tool - supporting better resource allocation and food security planning.
5.	Business Model (Revenue Model)	Freemium model: a free public dashboard (Tableau Public) for farmers, students and researchers, with a premium tier for state agriculture departments and agribusinesses offering advanced analytics, custom reports and data-feed access; also viable as an analytics-as-a-service offering for agri-tech firms and NGOs.
6.	Scalability of the Solution	Built on Tableau Data Extracts with incremental refresh, the solution can scale to real-time feeds from data.gov.in/Agmarknet, incorporate additional parameters (rainfall, soil health, market prices), and be replicated for other states or countries' agricultural datasets without redesign.